

H0 — decision summary (1/2)

llama31-8b (ctx 131,072, 1,024 heads)

INPUT VALIDITY PASS haystack real

task-level needle retrieved in 6/6 niah prompts (PASS, need 50%) [measurement]

head-level needle mass: max 0.8572 p99 0.2636 median 0.000205 (56/1024 heads >= 0.05)

heads above 0.001:363 0.005:239 0.01:174 0.05:56 0.1:30 0.25:11 [ADVISORY - MIN_MASS uncalibrated, not enforced]

RETIRED median-ladder gate (not enforced): niah 4.109 b cont 4.448 b delta -0.352 b 14.6% of heads niah>cont

HEADS IN BAND 12.8% (gain over best corner >= 2.0x at 3 b/token)

ROUTED GAIN 1.25x (geometric mean of max(gain,1) over all heads)

in-band heads median 2.7x p90 4.3x

ladder width median 4.27 b IQR [3.77, 4.74] 100% of heads above 1.4

tau median 2.96 excl. sinks 2.95

top-1 weight median 0.4071 95% mass in 1150 tokens (1.0% of ctx)

B=2b vs uniform 68.5x vs eviction 1.2x vs BEST corner 1.2x evicts 67%

B=3b vs uniform 33.6x vs eviction 1.0x vs BEST corner 1.0x evicts 52%

quantizer alpha @2b 0.943 (1.0 = no temperature distortion); spearman@2b 0.612 [H2 answered]

heads where eviction beats a 1-bit tier: 99%

heads where eviction beats a 2-bit tier: 54%

linearization check (predicted/measured gain @3b): 0.87 (1.0 = exact)

VERDICT: STOP – Only 13% of heads sit in the productive band and routing buys 1.25x overall. Uniform quantization or eviction already captures the gain almost everywhere; SIEVE would be a reframing without a result.

llama33-70b (ctx 131,072, 5,120 heads)

INPUT VALIDITY PASS haystack real

task-level needle retrieved in 3/4 niah prompts (PASS, need 50%) [measurement]

head-level needle mass: max 0.6931 p99 0.0478 median 0.000042 (51/5120 heads >= 0.05)

heads above 0.001:583 0.005:236 0.01:146 0.05:51 0.1:27 0.25:8 [ADVISORY - MIN_MASS uncalibrated, not enforced]

RETIRED median-ladder gate (not enforced): niah 4.713 b cont 5.223 b delta -0.379 b 12.3% of heads niah>cont

HEADS IN BAND 18.0% (gain over best corner >= 2.0x at 3 b/token)

ROUTED GAIN 1.34x (geometric mean of max(gain,1) over all heads)

in-band heads median 3.0x p90 5.1x

ladder width median 4.84 b IQR [3.82, 5.78] 99% of heads above 1.4

tau median 3.34 excl. sinks 3.34

top-1 weight median 0.1495 95% mass in 4808 tokens (4.0% of ctx)

B=2b vs uniform 76.7x vs eviction 1.4x vs BEST corner 1.4x evicts 68%

B=3b vs uniform 36.9x vs eviction 1.0x vs BEST corner 1.0x evicts 54%

quantizer alpha @2b 0.943 (1.0 = no temperature distortion); spearman@2b 0.552 [H2 answered]

heads where eviction beats a 1-bit tier: 99%

heads where eviction beats a 2-bit tier: 41%

linearization check (predicted/measured gain @3b): 0.80 (1.0 = exact)

VERDICT: NARROW – 18% of heads are in band (1.34x routed). Real but partial. Ship the router as part of the method and make the regime boundary a contribution.

mistral-7b (ctx 32,768, 1,024 heads)

INPUT VALIDITY PASS haystack real

task-level needle retrieved in 6/6 niah prompts (PASS, need 50%) [measurement]

head-level needle mass: max 0.9474 p99 0.6918 median 0.001416 (142/1024 heads >= 0.05)

heads above 0.001:568 0.005:367 0.01:287 0.05:142 0.1:100 0.25:44 [ADVISORY - MIN_MASS uncalibrated, not enforced]

RETIRED median-ladder gate (not enforced): niah 3.050 b cont 3.389 b delta -0.294 b 19.5% of heads niah>cont

HEADS IN BAND 41.4% (gain over best corner >= 2.0x at 3 b/token)

ROUTED GAIN 1.82x (geometric mean of max(gain,1) over all heads)

in-band heads median 3.0x p90 4.5x

ladder width median 3.22 b IQR [2.79, 3.69] 98% of heads above 1.4

tau median 2.21 excl. sinks 2.20

top-1 weight median 0.4166 95% mass in 1389 tokens (4.6% of ctx)

B=2b vs uniform 40.8x vs eviction 2.0x vs BEST corner 2.0x evicts 63%

B=3b vs uniform 25.4x vs eviction 1.8x vs BEST corner 1.7x evicts 48%

quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.619 [H2 answered]

heads where eviction beats a 1-bit tier: 98%

H0 — decision summary (2/2)

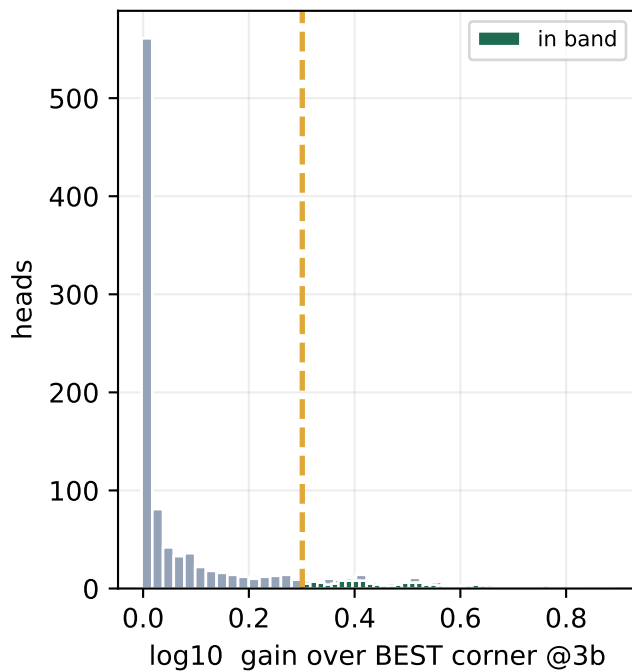
qwen15-moe-a2.7b (ctx 32,768, 384 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 2/6 niah prompts (FAIL, need 50%) [measurement]
head-level needle mass: max 0.9243 p99 0.3916 median 0.000189 (34/384 heads >= 0.05)
heads above 0.001:147 0.005:107 0.01:81 0.05:34 0.1:23 0.25:9 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 4.295 b cont 4.756 b delta -0.268 b 16.7% of heads niah>cont
HEADS IN BAND 20.8% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.39x (geometric mean of max(gain,1) over all heads)
in-band heads median 3.4x p90 5.5x
ladder width median 4.42 b IQR [3.56, 5.47] 99% of heads above 1.4
tau median 3.05 excl. sinks 3.05
top-1 weight median 0.3358 95% mass in 601 tokens (2.0% of ctx)
B=2b vs uniform 55.9x vs eviction 1.2x vs BEST corner 1.2x evicts 66%
B=3b vs uniform 30.4x vs eviction 1.0x vs BEST corner 1.0x evicts 52%
quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.632 [H2 answered]
heads where eviction beats a 1-bit tier: 96%
heads where eviction beats a 2-bit tier: 36%
linearization check (predicted/measured gain @3b): 0.96 (1.0 = exact)
VERDICT: NARROW – 21% of heads are in band (1.39x routed). Real but partial. Ship the router as part of the method and make the regime boundary a contribution.

qwen3-30b-a3b-2507 (ctx 131,072, 1,536 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 2/4 niah prompts (PASS, need 50%) [measurement]
head-level needle mass: max 0.9289 p99 0.5178 median 0.000387 (188/1536 heads >= 0.05)
heads above 0.001:647 0.005:455 0.01:375 0.05:188 0.1:114 0.25:52 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 4.119 b cont 4.572 b delta -0.382 b 16.9% of heads niah>cont
HEADS IN BAND 4.1% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.11x (geometric mean of max(gain,1) over all heads)
in-band heads median 2.3x p90 3.0x
ladder width median 4.20 b IQR [3.66, 5.25] 100% of heads above 1.4
tau median 2.90 excl. sinks 2.90
top-1 weight median 0.4632 95% mass in 626 tokens (0.5% of ctx)
B=2b vs uniform 53.4x vs eviction 1.0x vs BEST corner 1.0x evicts 68%
B=3b vs uniform 29.2x vs eviction 1.0x vs BEST corner 1.0x evicts 54%
quantizer alpha @2b 0.942 (1.0 = no temperature distortion); spearman@2b 0.521 [H2 answered]
heads where eviction beats a 1-bit tier: 100%
heads where eviction beats a 2-bit tier: 73%
linearization check (predicted/measured gain @3b): 0.99 (1.0 = exact)
VERDICT: STOP – Only 4% of heads sit in the productive band and routing buys 1.11x overall. Uniform quantization or eviction already captures the gain almost everywhere; SIEVE would be a reframing without a result.

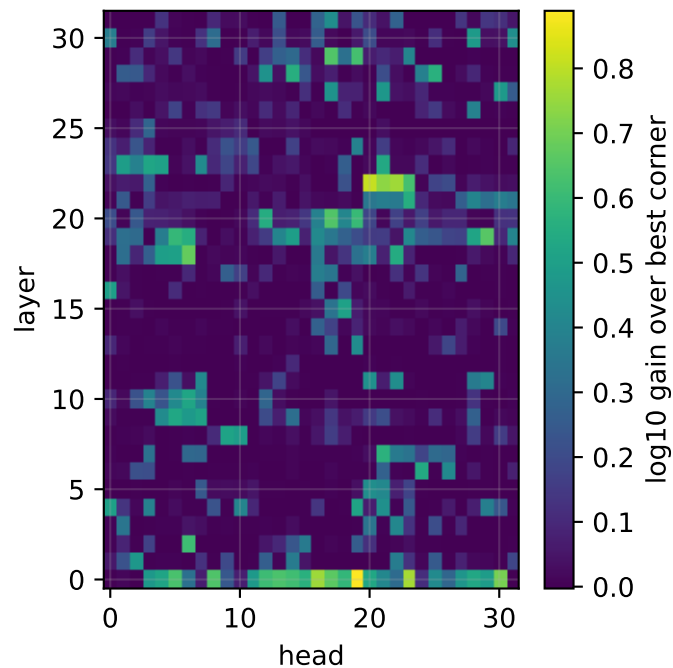
qwen3-8b (ctx 40,960, 1,152 heads)
INPUT VALIDITY PASS haystack real
task-level needle retrieved in 6/6 niah prompts (PASS, need 50%) [measurement]
head-level needle mass: max 0.9498 p99 0.7034 median 0.000370 (176/1152 heads >= 0.05)
heads above 0.001:491 0.005:401 0.01:336 0.05:176 0.1:114 0.25:60 [ADVISORY - MIN_MASS uncalibrated, not enforced]
RETIRED median-ladder gate (not enforced): niah 4.488 b cont 5.061 b delta -0.432 b 22.5% of heads niah>cont
HEADS IN BAND 5.6% (gain over best corner >= 2.0x at 3 b/token)
ROUTED GAIN 1.13x (geometric mean of max(gain,1) over all heads)
in-band heads median 2.4x p90 2.9x
ladder width median 4.65 b IQR [3.96, 6.15] 100% of heads above 1.4
tau median 3.21 excl. sinks 3.20
top-1 weight median 0.1813 95% mass in 413 tokens (1.1% of ctx)
B=2b vs uniform 53.3x vs eviction 1.0x vs BEST corner 1.0x evicts 67%
B=3b vs uniform 29.9x vs eviction 1.0x vs BEST corner 1.0x evicts 53%
quantizer alpha @2b 0.940 (1.0 = no temperature distortion); spearman@2b 0.628 [H2 answered]
heads where eviction beats a 1-bit tier: 100%

llama31-8b — ctx 131,072

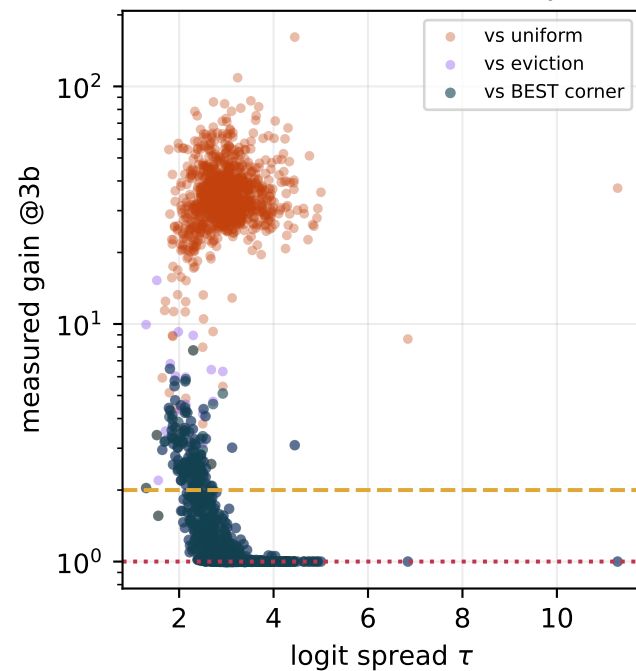
HEADS IN BAND: 13%



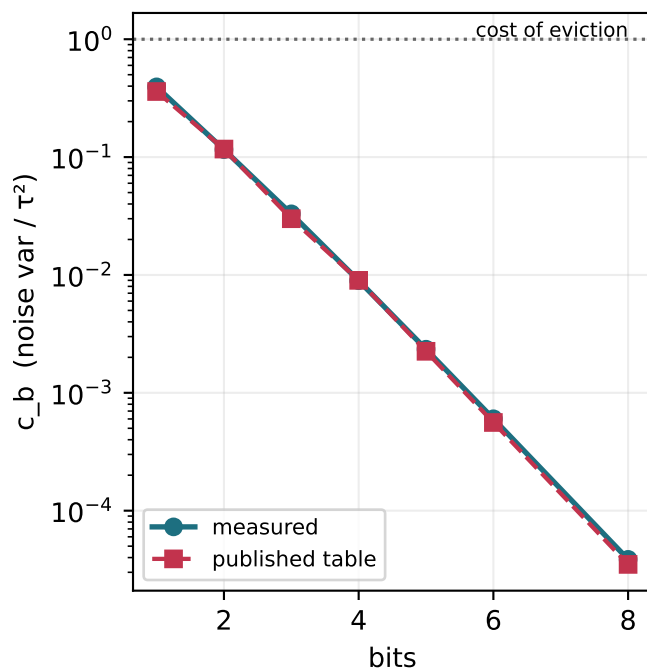
Which heads the router sends to SIEVE



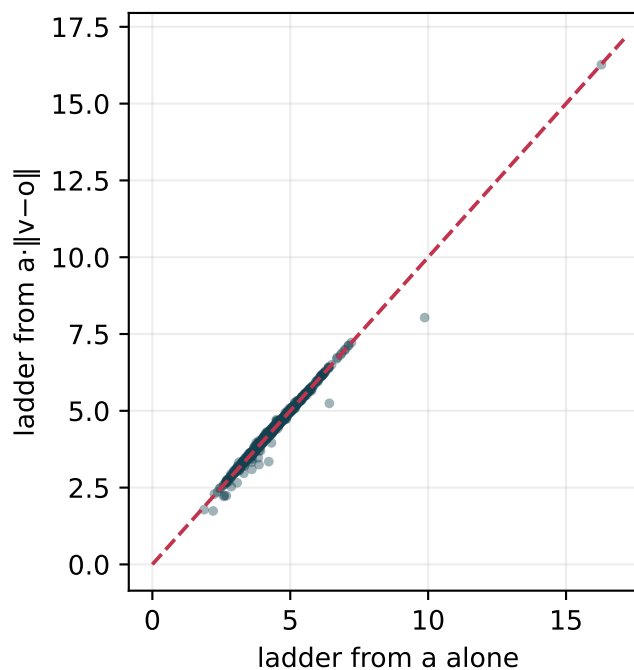
Where does the interior actually win?



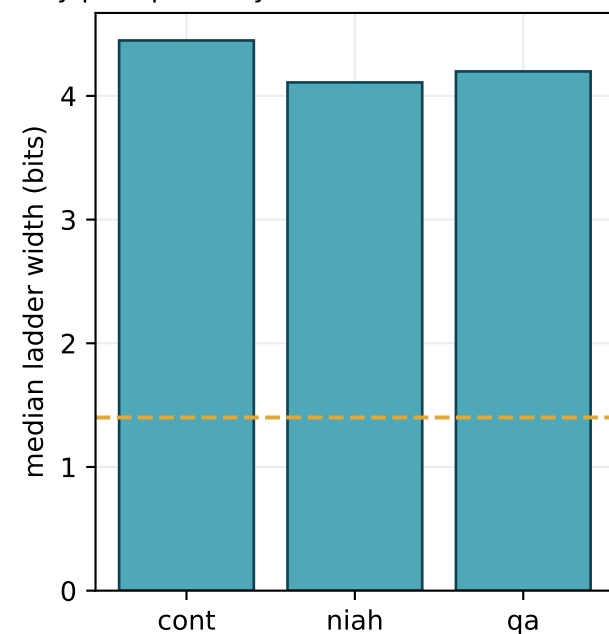
Measured vs assumed noise



Does the value term matter?

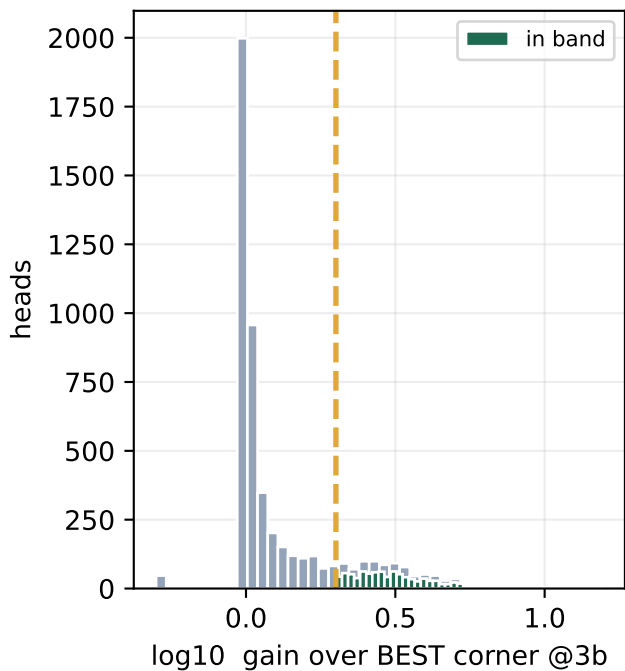


By prompt family — niah—cont -0.352 b, PA

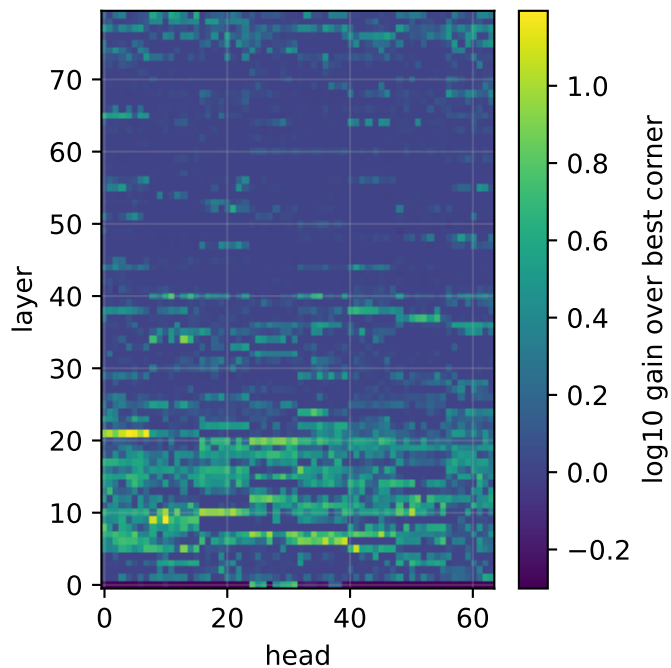


llama33-70b — ctx 131,072

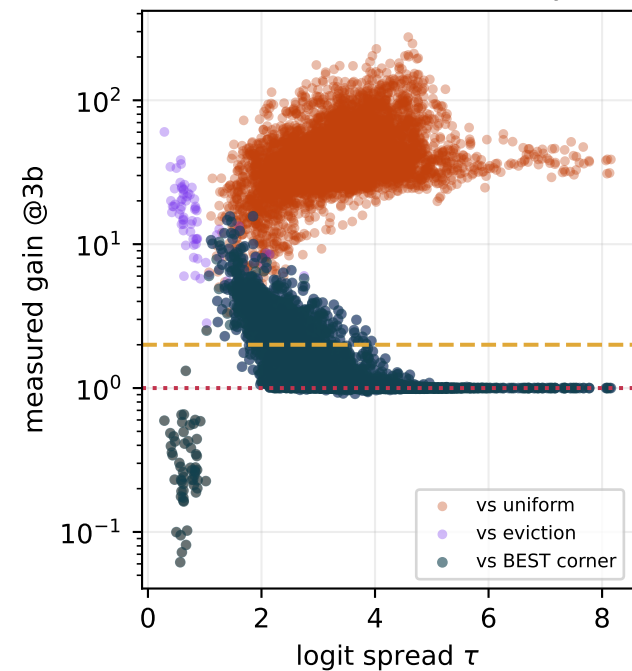
HEADS IN BAND: 18%



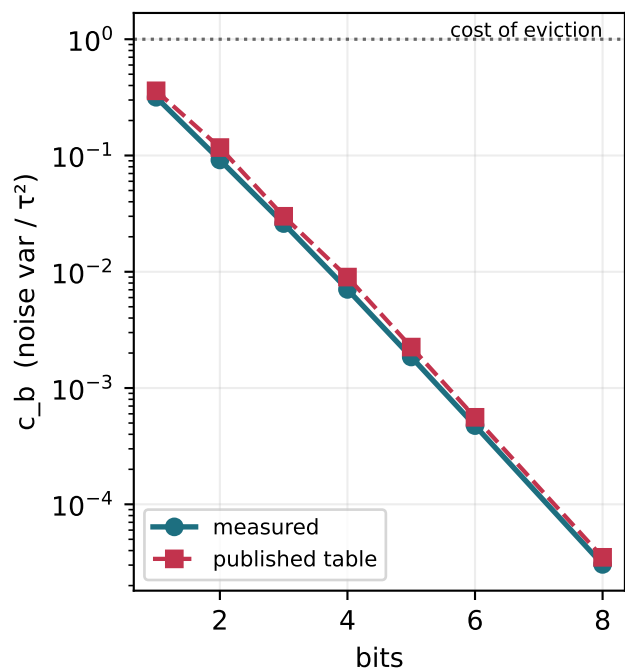
Which heads the router sends to SIEVE



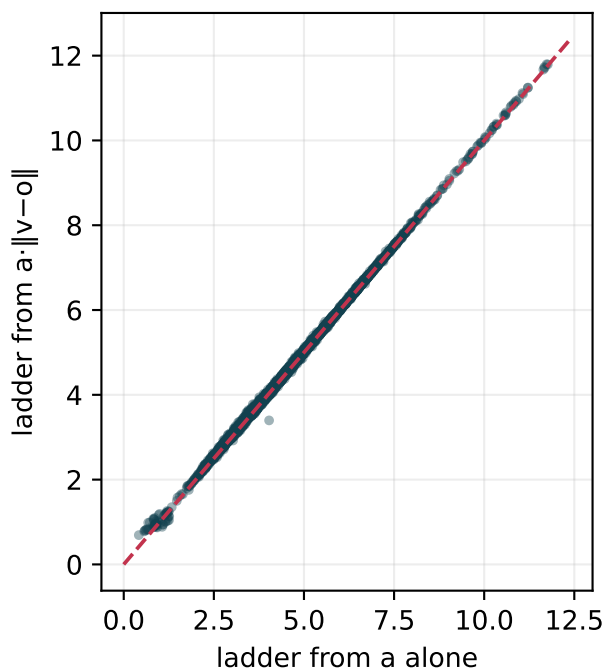
Where does the interior actually win?



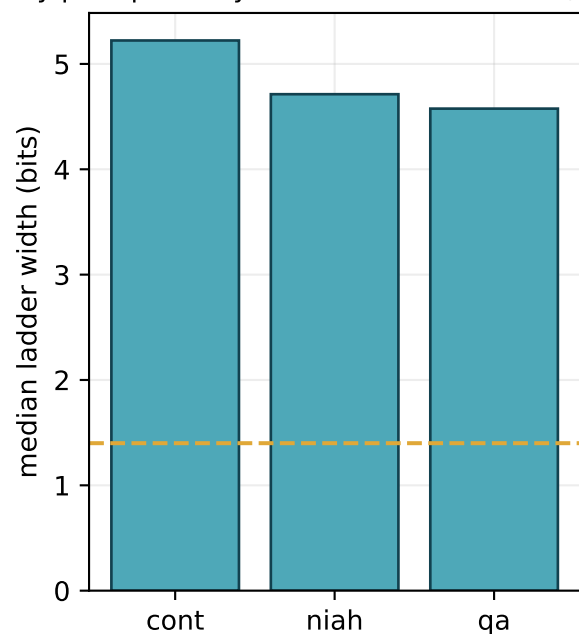
Measured vs assumed noise



Does the value term matter?

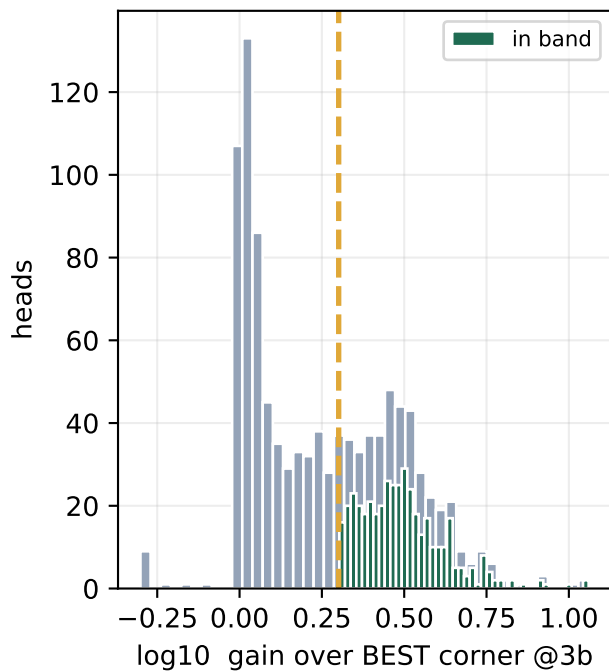


By prompt family — niah—cont -0.379 b, PA

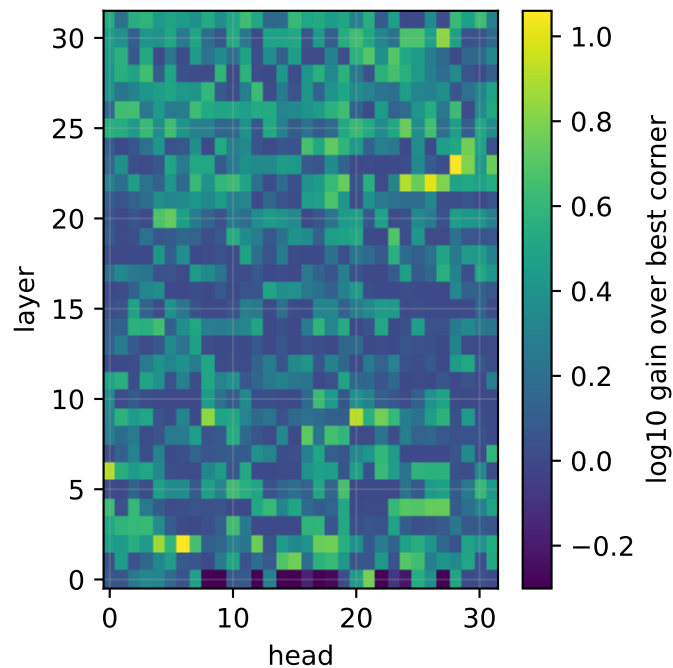


mistral-7b — ctx 32,768

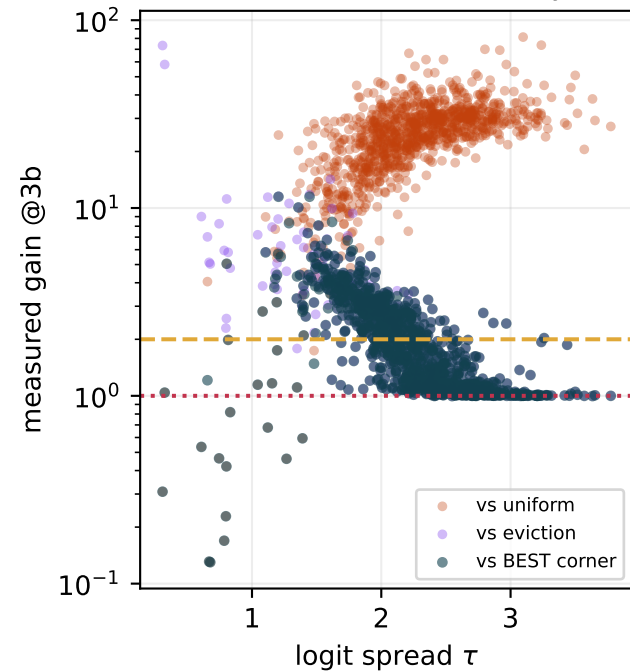
HEADS IN BAND: 41%



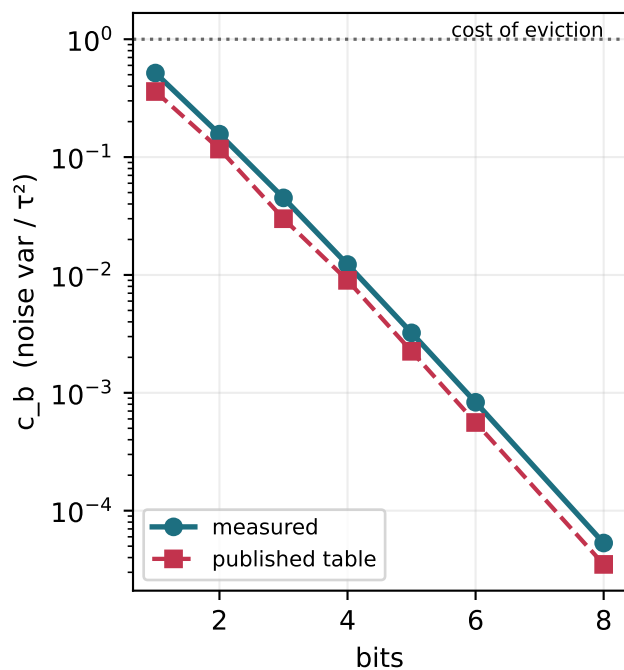
Which heads the router sends to SIEVE



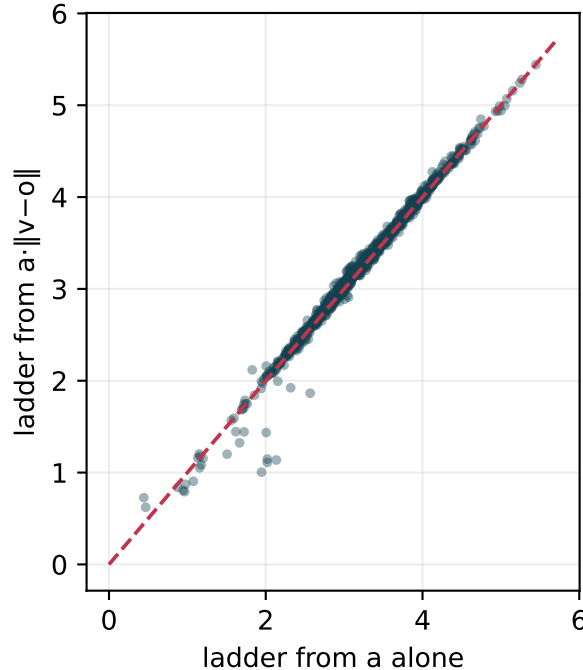
Where does the interior actually win?



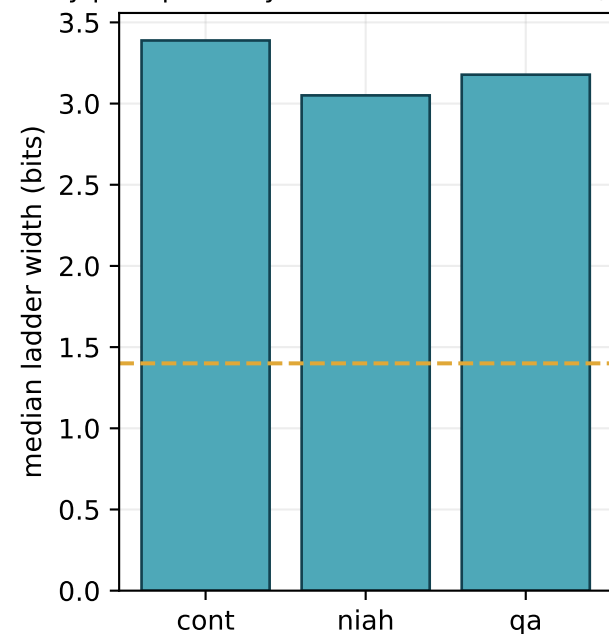
Measured vs assumed noise



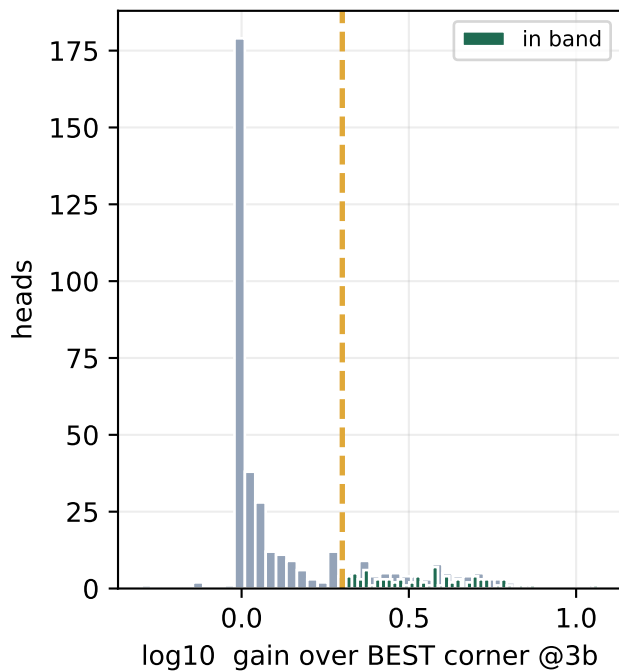
Does the value term matter?



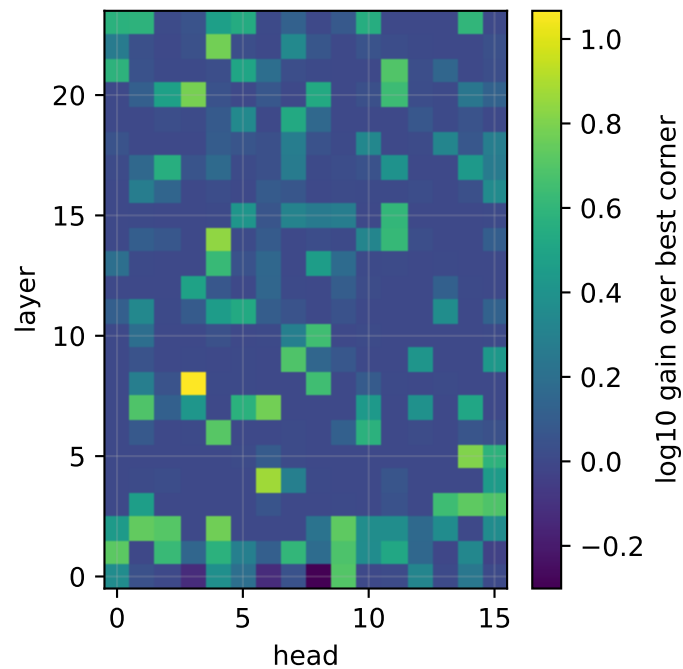
By prompt family — niah-cont -0.294 b, PA



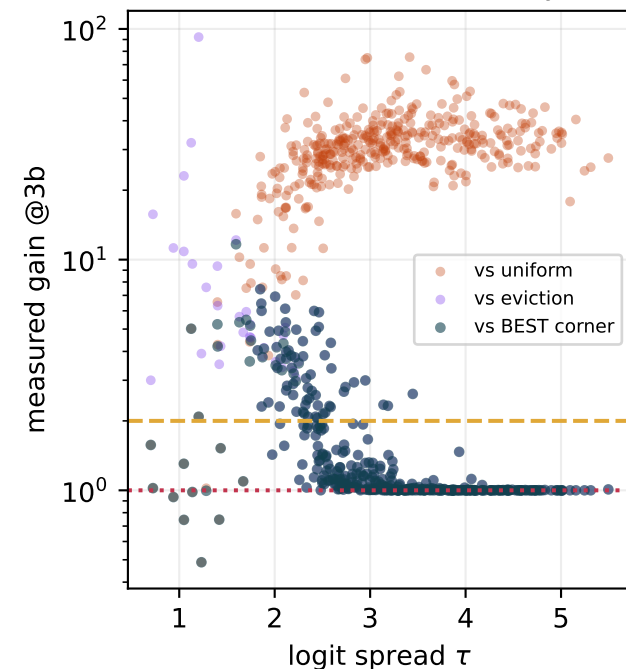
HEADS IN BAND: 21%



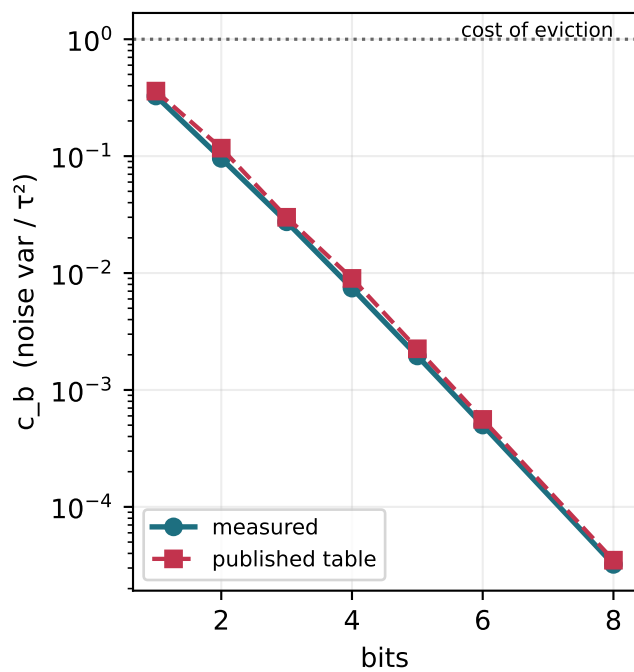
Which heads the router sends to SIEVE



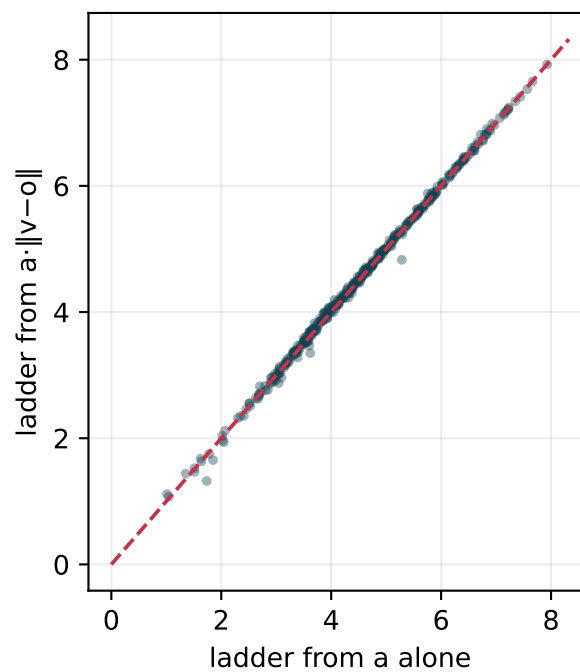
Where does the interior actually win?



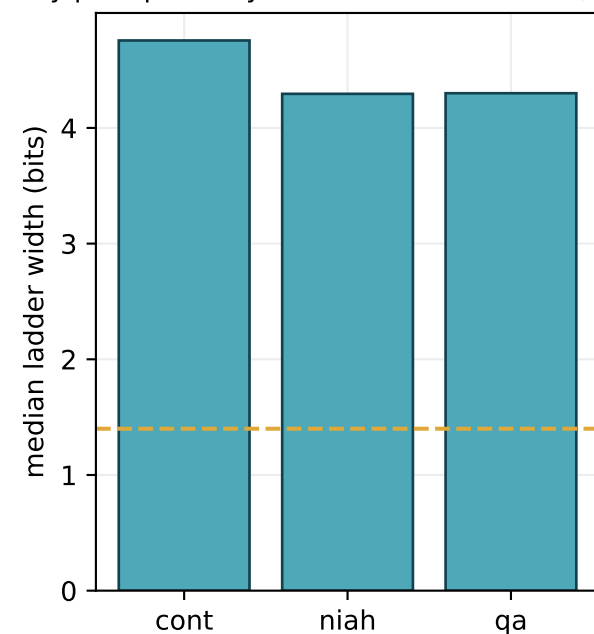
Measured vs assumed noise



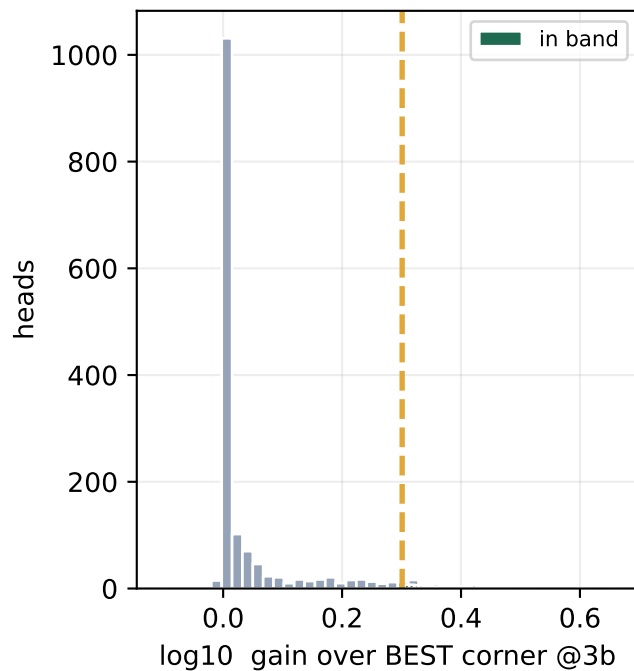
Does the value term matter?



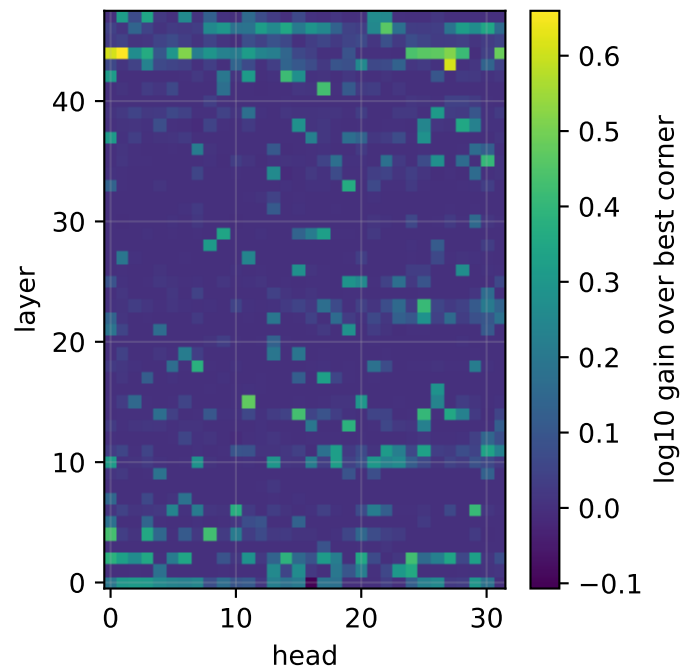
By prompt family — niah—cont -0.268 b, PA



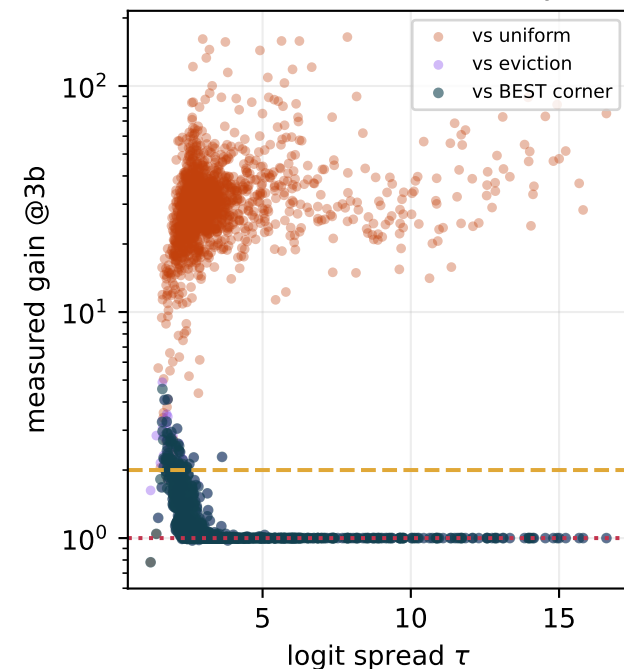
HEADS IN BAND: 4%



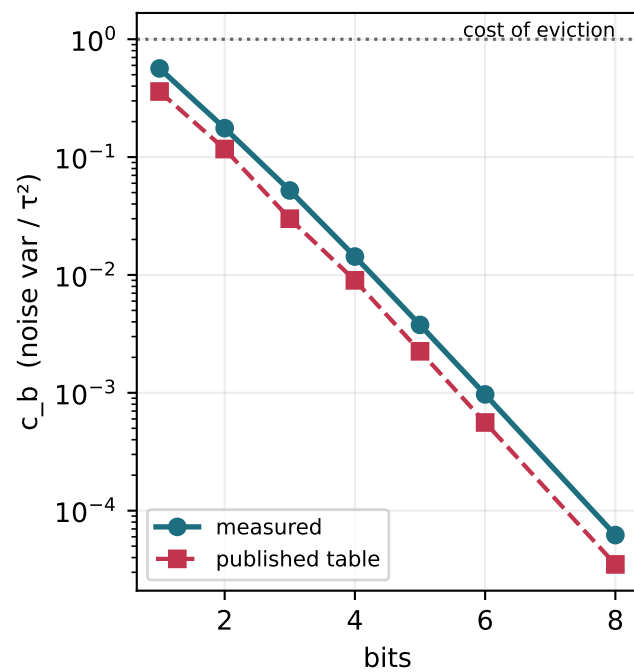
Which heads the router sends to SIEVE



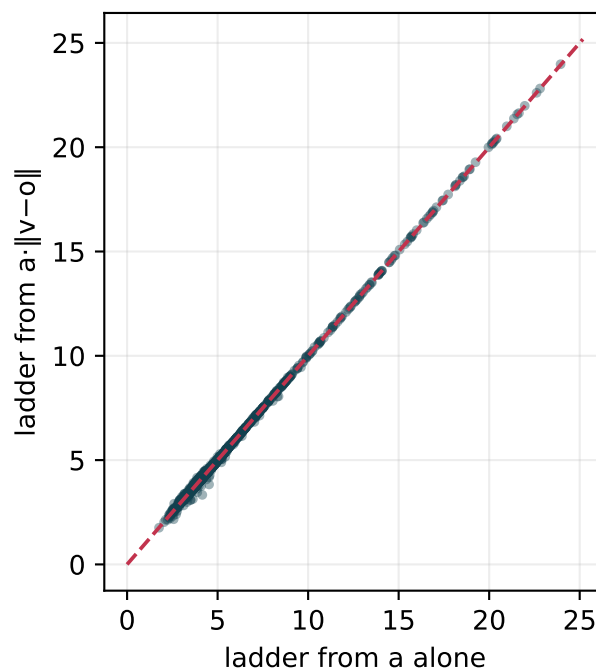
Where does the interior actually win?



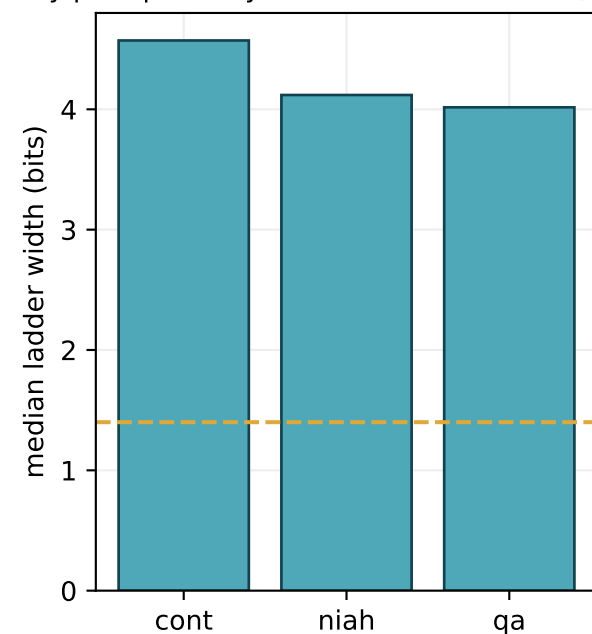
Measured vs assumed noise



Does the value term matter?

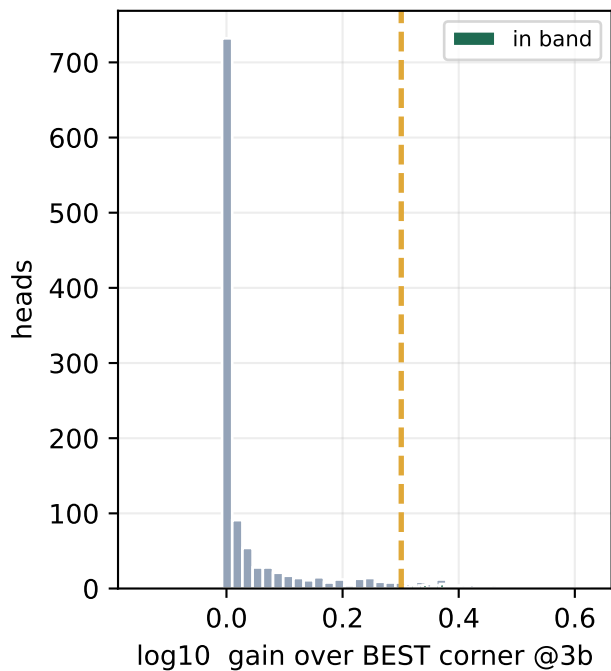


By prompt family — niah-cont -0.382 b, PA

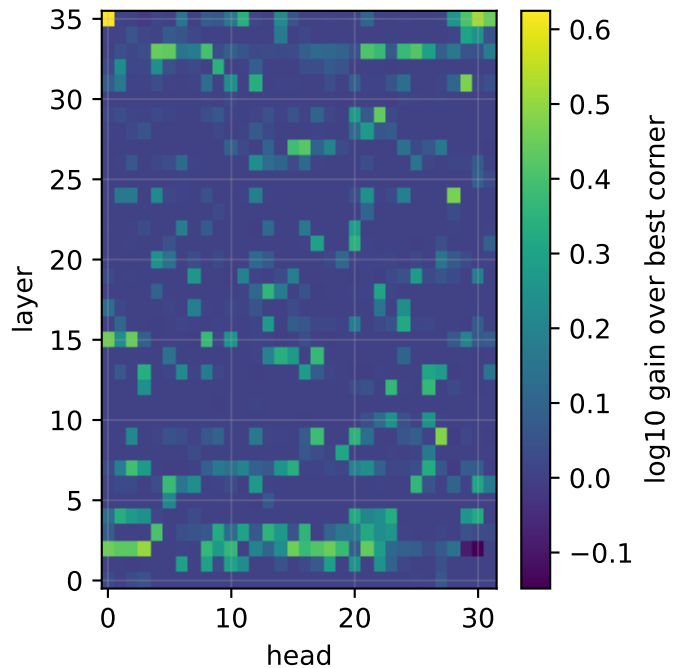


qwen3-8b — ctx 40,960

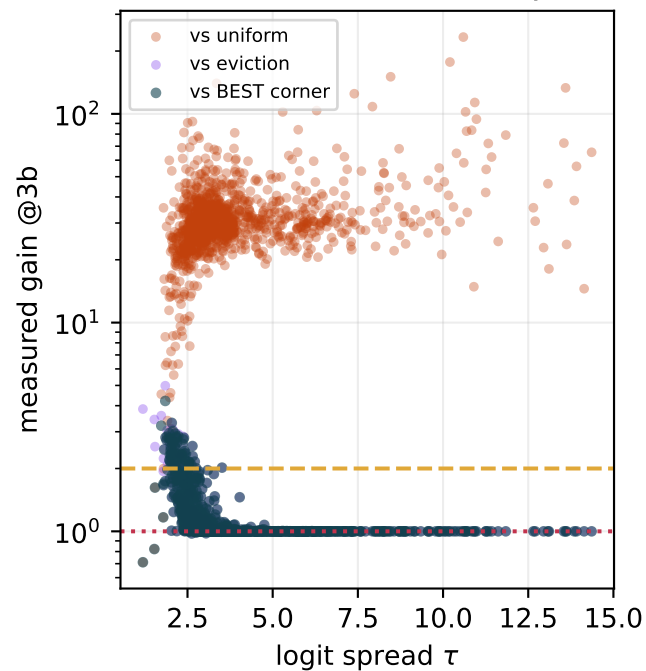
HEADS IN BAND: 6%



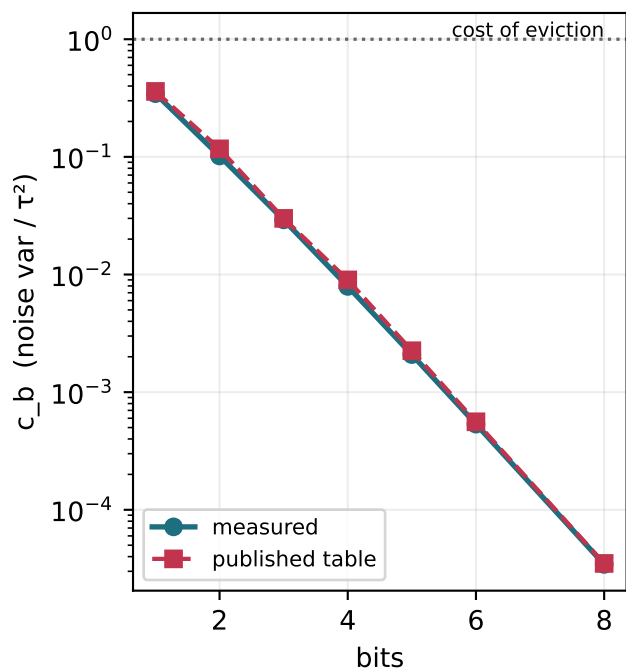
Which heads the router sends to SIEVE



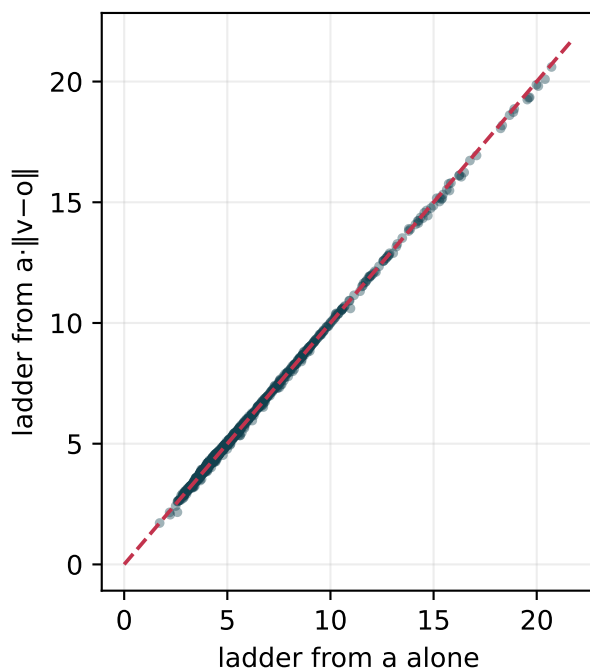
Where does the interior actually win?



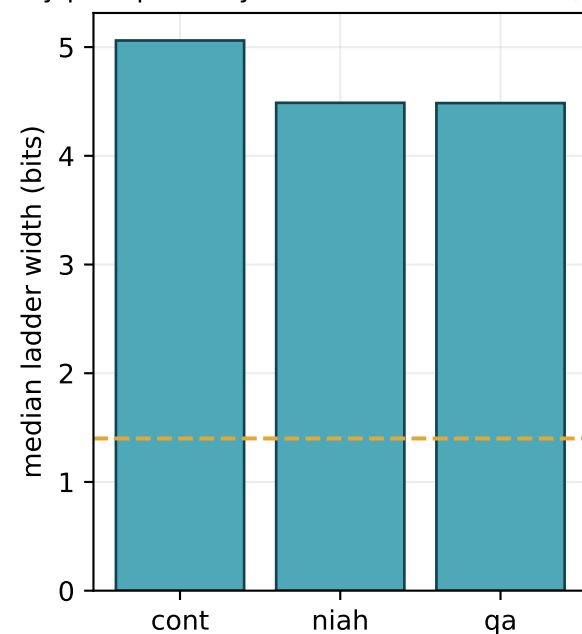
Measured vs assumed noise



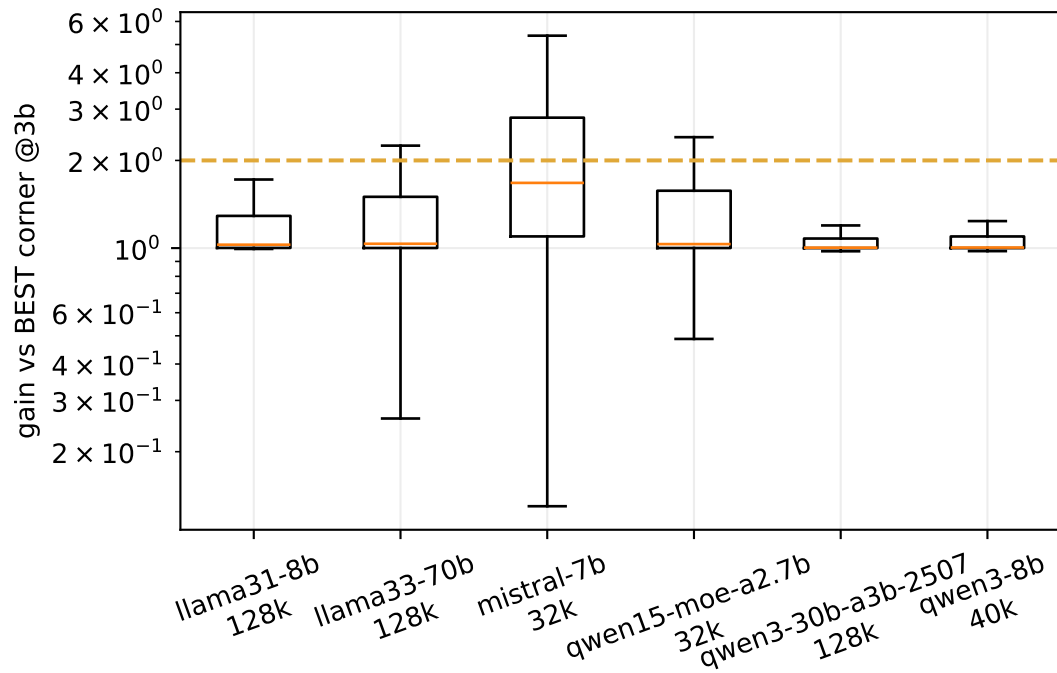
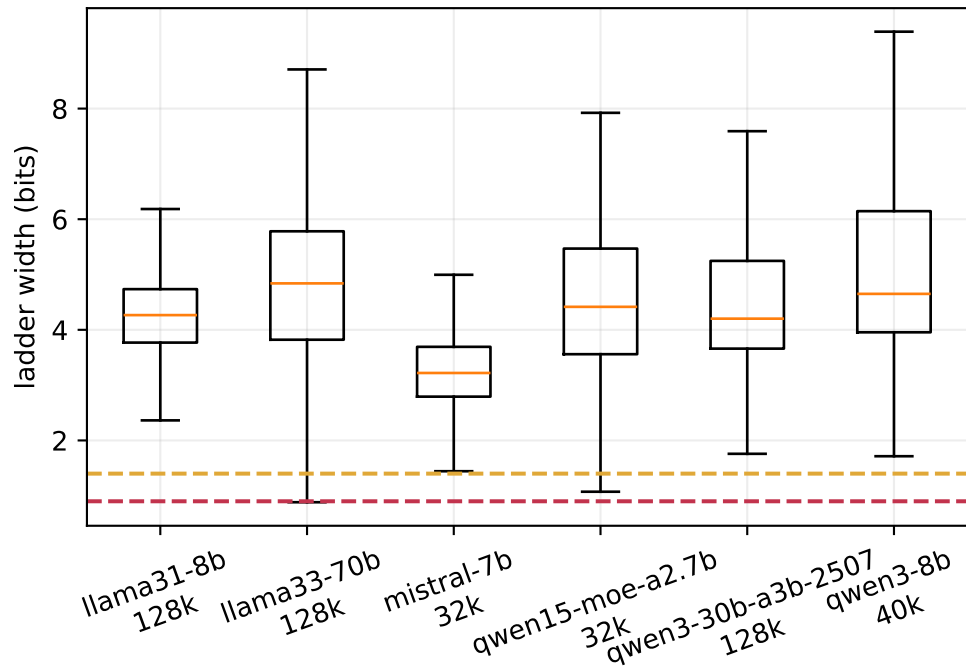
Does the value term matter?



By prompt family — niah—cont -0.432 b, PAS



Cross-model comparison



Phase diagram on derived axes ($\phi = n_{95}/L$ retired: it divides by context length)

